

Scheme of Valuation/Answer Key			
(Scheme of evaluation (marks in brackets) and answers of problems/key)			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
Seventh Semester B.Tech Degree (R, S) Examination November 2024 (2019 Scheme)			
Course Code: CXT401			
Course Name: USER INTERFACE SOFTWARE AND TECHNOLOGY (UIST)			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks.	Marks
1		(Definition 3 marks) I want to accomplish something now, not later. Instant gratification in UIST refers to the user’s desire for immediate rewards or feedback when interacting with a system or interface. • If someone starts using an application and gets a “success experience” within the first few seconds, that’s gratifying! He’ll be more likely to keep using it, even if it gets harder later. • They will feel more confident in the application, and more confident in themselves, than if it had taken a while to figure things out. Importance: • User Satisfaction: Providing quick responses and results enhances the user experience and keeps users engaged. • Efficiency: Users prefer interfaces that offer immediate feedback, which helps them feel productive and reduces frustration.	(3)
2		• Easy to use: Anybody with just basic knowledge of HTML and CSS can start using Bootstrap • Responsive features: Bootstrap's responsive CSS adjusts to phones, tablets, and desktops • Mobile-first approach: In Bootstrap, mobile-first styles are part of the core framework • Browser compatibility: Bootstrap 5 is compatible with all modern browsers (Chrome, Firefox, Edge, Safari, and Opera). Note that if you need support for IE11 and down, you must use either BS4 or BS3.	(3)

3		(Definition 1 mark + importance 2 marks) Information architecture (IA) is the art of organizing an information space. It encompasses many things: presenting, searching, browsing, labeling, categorizing, sorting, manipulating, and strategically hiding information.	(3)
4		(Definition 1 mark +code 2 marks) Cards support a wide variety of content, including images, text, list groups, links, and more. <u>Body</u> The building block of a card is the .card-body. Use it whenever you need a padded section within a card. <pre><div class="card"> <div class="card-body"> This is some text within a card body. </div> </div></pre>	(3)
5		Definition or code give 3 marks A "navbar" is an area on a page that contains navigation components (links, buttons, etc) for getting to other pages of the website. With Bootstrap, a navigation bar can extend or collapse, depending on the screen size. A standard navigation bar is created with the .navbar class, followed by a responsive collapsing class: .navbar-expand-xxl xl lg md sm. To add links inside the navbar, use either an element (or a <div>) with class="navbar-nav". Then add elements with a .nav-item class followed by an <a> element with a .nav-link class: Add the .active class on .nav-link to indicate the current page.	(3)
6		Explanation gives 3 marks Navigating an unfamiliar web page or application window incurs a cognitive cost (the amount of information needed)	(3)

		Users quickly assess the layout, content, and navigation options, which requires mental effort and time. Even familiar interfaces can impose this cognitive load, compounded by actual delays in loading or displaying content. This cost impacts user experience, as prolonged loading times can frustrate users, leading them to abandon pages or avoid exploring the site altogether. Effective design minimizes cognitive and loading costs to retain and engage users more effectively.	
7		Definition -3 Marks When modules are not essential, they remain closed by default to avoid clutter. When they are not essential enough to be open by default. When modules are not needed, collapsing them optimizes space for the main content. When users interact with the page, the importance of modules varies, as some find specific ones more useful than others. When using Collapsible Panels, they are independent and do not suggest relationships, unlike Tabs or Accordions.	(3)
8		Definition -3 Marks Stacks modules of content vertically. Allows independent opening/closing of each module. Ideal for heterogeneous content such as text blocks, lists, images, etc. User-friendly for limited space pages	(3)
9		Definition or code 3 marks Textarea: <code><textarea class="form-control" rows="3" placeholder="Enter text"></textarea></code> • Multi-line input for longer text entries. Adjustable in size using the rows attribute. • Select Menus: <code><select class="form-select" aria-label="Select option"></code> <code><option selected>Select option</option></code> <code><option value="1">Option 1</option></code>	(3)

		<option value="2">Option 2</option> </select> Drop-down menu for selecting one or more options from a list.	
10		Any 3 points give 3 marks User-Centric Design: Always design actions and commands with the user's needs and expectations in mind. Ensure they are intuitive and align with user workflows. Error Handling: Implement clear error messages and recovery options to guide users if an action or command fails. Accessibility: Ensure that interactive elements are accessible to all users, including those with disabilities, by supporting keyboard navigation, screen readers, and other assistive technologies.	(3)
PART B			
<i>Answer one full question from each module, each carries 14 marks.</i>			
		Module I	
11	a)	Users' Motivation When you carry on a conversation with someone about a given subject, you adjust what you say according to your understanding of the other person. You might consider ➤ How much he cares about the subject, ➤ How much he already knows about it, ➤ How receptive he is ➤ whether he's even interested in the conversation in the first place If you get any of that wrong, bad things happen he might feel uninterested, impatient, or utterly baffled. (about user motivation give 3 marks) PATTERNS Safe exploration Instant gratification Streamlined Repetition Satisficing Habituation	(8)

		For each pattern (with explanation and example) give 1 mark(5*1=5)	
	b)	Containers(3 types of containers, explanation about each container give 2 marks) - Containers are the most basic layout element in Bootstrap and are required when using our default grid system. - Containers are used to contain, pad, and (sometimes) center the content within them. While containers <i>can</i> be nested, most layouts do not require a nested container. Bootstrap comes with three different containers: <code>.container</code>, which sets a <code>max-width</code> at each responsive breakpoint <code>.container-fluid</code>, which is <code>width: 100%</code> at all breakpoints <code>.container-{breakpoint}</code>, which is <code>width: 100%</code> until the specified breakpoint	(6)
		OR	
12	a)	Habituation (give 3 marks) That gesture works everywhere else; why doesn't it work here, too?" - Habituation refers to the process where users become familiar with certain actions or gestures in a user interface through repeated use. Benefits: Expertise: Users become more skilled and efficient with the tool. Flow: Helps users work smoothly without needing to think about every action. Risks: Unexpected Problems: Users might rely on habitual gestures that don't work in all situations. Confusion: If a habit doesn't work or causes issues, users may be confused and need to figure out how to fix it or correct mistakes. Understanding habituation helps in designing interfaces that support efficient use while minimizing potential problems	(6)

	Streamlined Repetition ((give 3 marks) Streamlined repetition involves making repetitive tasks more efficient and less tedious by simplifying and automating them. Importance: • Increases Efficiency: Reduces the time and effort required to perform repeated actions. • Enhances User Experience: Makes interactions smoother and less frustrating for users. Design Strategies: • Automate Tasks: Implement features like shortcuts, macros, or automation tools to handle repetitive actions. • Consistent Design: Use consistent layouts and controls to make repeated tasks easier to perform. • Provide Feedback: Offer clear feedback for repeated actions to confirm that tasks are completed successfully. Designing for streamlined repetition helps users perform repetitive tasks more quickly and with less effort, improving overall efficiency and satisfaction. EXAMPLE Find and Replace dialog boxes, commonly used in text editors (like Word or email composers), are designed to simplify repetitive tasks. Users enter the old phrase and the new phrase, then use a single Replace button to change each occurrence one by one. If users are sure they want to replace every instance, they can click the Replace All button, which performs the entire task with just one click.	
b)	Spacing (Give 5 marks)	(8)

Spacing

Bootstrap 5 has a wide range of responsive margin and padding utility classes. They work for all breakpoints: `xs` ($\leq 576\text{px}$), `sm` ($> 576\text{px}$), `md` ($\geq 768\text{px}$), `lg` ($\geq 992\text{px}$), `xl` ($\geq 1200\text{px}$) or `xxl` ($\geq 1400\text{px}$):

The classes are used in the format: `{property}{sides}-{size}` for `xs` and `{property}{sides}-{breakpoint}-{size}` for `sm`, `md`, `lg`, `xl` and `xxl`.

Where *property* is one of:

- `m` - sets `margin`
- `p` - sets `padding`

Where *sides* is one of:

- `t` - sets `margin-top` or `padding-top`
- `b` - sets `margin-bottom` or `padding-bottom`
- `s` - sets `margin-left` or `padding-left`
- `e` - sets `margin-right` or `padding-right`
- `x` - sets both `padding-left` and `padding-right` or `margin-left` and `margin-right`
- `y` - sets both `padding-top` and `padding-bottom` or `margin-top` and `margin-bottom`
- blank - sets a `margin` or `padding` on all 4 sides of the element

Where *size* is one of:

- `0` - sets `margin` or `padding` to `0`
- `1` - sets `margin` or `padding` to `.25rem`
- `2` - sets `margin` or `padding` to `.5rem`
- `3` - sets `margin` or `padding` to `1rem`
- `4` - sets `margin` or `padding` to `1.5rem`
- `5` - sets `margin` or `padding` to `3rem`
- `auto` - sets `margin` to `auto`

Breakpoints (Give 3 marks)

Breakpoint	Class infix	Dimensions
X-Small	<i>None</i>	$< 576\text{px}$
Small	<code>sm</code>	$\geq 576\text{px}$
Medium	<code>md</code>	$\geq 768\text{px}$
Large	<code>lg</code>	$\geq 992\text{px}$
Extra large	<code>xl</code>	$\geq 1200\text{px}$
Extra extra large	<code>xxl</code>	$\geq 1400\text{px}$

		Module II	
13	a)	(Explanation or code of each point give 2 marks) Image shapes Aligning images Responsive images Images come in all sizes. So do screens. Responsive images automatically adjust to fit the size of the screen. Images in Bootstrap are made responsive with .img-fluid. This applies max-width: 100%; and height: auto; to the image so that it scales with the parent element. <pre></pre> Image Shapes Float an image to the left with the .float-start class or to the right with .float-end <pre></pre> <pre></pre>	(6)
	b)	(Explanation about list give 2 marks + explanation about .list-group, .list-group-item, and .list-inline give 6 marks) List groups are a flexible and powerful component for displaying a series of content. Modify and extend them to support just about any content within. The most basic list group is an unordered list with list items: To create a basic list group, use an element with class .list-group, and elements with class .list-group-item: <pre><ul class="list-group"></pre> <pre><li class="list-group-item">First item</pre>	(8)

		<pre> <li class="list-group-item">Second item <li class="list-group-item">Third item </pre> The items in a list group can also be hyperlinks. This will add a grey background color on hover: <pre> <div class="list-group"> First item Second item Third item </div> </pre> To create a list group with linked items, use <div> instead of and <a> instead of list-inline Remove a list's bullets and apply some light margin with a combination of two classes, .list-inline and .list-inline-item. <pre> <ul class="list-inline"> <li class="list-inline-item">This is a list item. <li class="list-inline-item">And another one. <li class="list-inline-item">But they're displayed inline. </pre> Active and disabled state	
		OR	
14	a)	Detailed explanation gives 5 marks+ 2 marks for example Feature, search and browse (What, when, how and why) - Put three elements on the main page of the site or app: a featured article or product, a search box, and a list of items or categories that can be browsed - This approach helps users quickly find engaging content and encourages them to explore more of what your site has to offer. - Successful sites often use three key elements together: search ,browsing options, and featured items.	(7)

		➤ Search lets users find exactly what they want. ➤ while browsing helps them discover new things. ➤ Featured items grab attention with appealing images and headlines, giving users something interesting to engage with right away. • This approach can make users more interested in your site and keep them exploring. • A user who lands on a page now has something to read or experiment with, without doing any additional work at all and he may find it more interesting than whatever he originally came for.	
	b)	Detailed explanation gives 5 marks+ 2 marks for example News Stream (What, when, how and why) • Display, time-sensitive items in a reverse chronological order, with the latest at the top. • Update the list dynamically and combine items from various sources into a single stream. • Your site or app uses communication channels like blogs, email, social media, or news updates to deliver timely content.	(7)
		Module III	
15	a)	Detailed explanation gives 5 marks+ 1.5 marks for each example(1.5 *2=3) Clear entry points (What, when, how and why) Present only a few main entry points into the interface; make them task-oriented and descriptive. Use when You're designing a site or app that has a lot of first-time or infrequent users. Most of these users would be best served by reading a certain piece of introductory text, doing an initial task, or choosing from a very small number of frequently used options. If the purpose is clear to nearly everyone who starts using it, and if adding another navigation step might annoy most users (especially those who are intermediate to expert), this design choice might not be the best.	(8)

	b)	Definition of navigation and navigation design 2 marks + ways to navigate 4 marks NAVIGATION In terms of User Interface (UI) design, Navigation is defined as the set of actions guiding the users navigate/move around the app so that they are able to complete their tasks easily. NAVIGATION DESIGN Navigation design involves creating, analyzing, and implementing ways for users to navigate through a website or app. Effective navigation design is central to good user interface as it directly impacts user interaction and product use. HOW TO NAVIGATE SIGNPOSTS WAYFINDING ➤ Good signage ➤ Environmental clues ➤ Maps	(6)
		OR	
16	a)	Definition or code of dropdown – 4 marks + 6 marks for its subcomponents A dropdown menu is a toggleable menu that allows the user to choose one value from a predefined list: <pre><divclass="dropdown"> <button type="button" class="btn btn-primary dropdown-toggle" data-bs-toggle="dropdown"> Dropdown button </button> <ul class="dropdown-menu"> Link 1 Link 2 Link 3 </div></pre>	(10)

		The .dropdown class indicates a dropdown menu. To open the dropdown menu, use a button or a link with a class of .dropdown-toggle and the data-bs-toggle="dropdown" attribute. data-bs-toggle: This attribute is used to describe what type of toggling will be done to the button to which it is attached. Add the .dropdown-menu class to a container element, like <code><div></code> or <code></code>, to actually build the dropdown menu. Then add the .dropdown-item class to each element (links or buttons) inside the dropdown menu. Dropdown Divider Dropdown Header Disable and Active items Sizing Dark dropdowns Directions Menu alignment Responsive alignment	
	b)	Correct code or Explanation -4 marks <pre><ul class="pagination"> <li class="page-item"> Previous <li class="page-item">1 <li class="page-item">2 <li class="page-item">3 <li class="page-item">Next </pre>	(4)
		Module IV	
17	a)	Diagonal balance (Explanation 3.5 Marks) - Page elements are in an asymmetric fashion, but balance it by using visual weight - For a Layout, a logo or header is at the top left, while a button or interactive link at the bottom right Liquid layout (Explanation 3.5 Marks) - Resizing the window automatically makes the contents react accordingly	(7)

		• More size means more contents can be shown, or existing contents are shown as larger • Shrinking size means less contents, or smaller icons are used	
	b)	Detailed explanation and advantages give 5 marks+ 2 marks for example Grid of Equals (What, when, how and why) Arrange content items in a grid or matrix. Each item should follow a common template, and each item's visual weight should be similar. Link to jump pages as necessary	(7)
		OR	
18	a)	Definition of page layout 2 marks+ each layout 2 marks (6*2=12) • Page layout is the art of manipulating the user's attention on a page to convey meaning, sequence, and points of interaction • several elements of page layout: ➤ visual hierarchy ➤ visual flow ➤ how to use dynamic displays. Layout patterns Centerstage grid of equals module tabs accordion collapsible panels diagonal balance liquid layout.	(14)
		Module V	
19	a)	Definition of action panels –1 mark + types of action panels 3 marks + design considerations 3 marks Action Panels Purpose: Action panels are dedicated sections or components of an interface that group related actions or commands together. They help users manage or interact with content or settings more efficiently. Types of Action Panels:	(7)

	• Sidebar Panels: Vertical panels that slide in from the side of the screen, typically used for navigation or managing settings. • Modal Panels: Overlay panels that appear on top of the main content, used for tasks like editing or detailed settings. • Floating Action Buttons (FABs): Prominently placed buttons that trigger primary actions, often used in mobile apps to provide quick access to core functionalities. <u>Design Considerations:</u> • Organization: Group related actions logically to minimize user effort and confusion. Use clear labels and visual hierarchy to guide users. • Visibility and Accessibility: Ensure action panels are easy to access and don't obstruct important content. Consider how they will be displayed on different devices or screen sizes. **Consistency • Consistency: Maintain consistent design patterns for action panels across the application to help users quickly learn and navigate the interface. Ensure the style, layout, and behavior of action panels align with the overall design language.	
b)	Explanation or code -7 marks Custom Form Elements Bootstrap 5 allows for extensive customization of form elements, providing a way to create more unique and tailored UI components. Custom Forms: Custom Checkboxes and Radio Buttons: <pre><div class="form-check"> <input class="form-check-input" type="checkbox" id="customCheck1"> <label class="form-check-label" for="customCheck1"> Check this custom checkbox </label> </div> <div class="form-check"></pre>	(7)

		<pre> <input class="form-check-input" type="radio" name="options" id="customRadio1" value="option1"> <label class="form-check-label" for="customRadio1"> Select this custom radio </label> </div> </pre>	
		OR	
20	a)	Explanation or code -7 marks Input Groups - Input groups allow you to prepend or append additional elements to form controls. This is useful for adding icons, text, or buttons inside the input field. - Basic Input Group: <pre> <div class="input-group"> \$ <input type="text" class="form-control" placeholder="Amount"> </div> </pre> Adds a text prefix or suffix to an input field, such as a currency symbol or unit of measurement. Input Group with Button: <pre> <div class="input-group"> <input type="text" class="form-control" placeholder="Enter something"> <button class="btn btn-outline-secondary" type="button">Button</button> </div> </pre>	(7)
	b)	Definition 2 marks + for each type give 1 mark (3*1=3) + Design Considerations: give 2 marks Buttons Purpose: Buttons are primary interactive elements used to initiate actions or commands. They are a core component of most user interfaces, facilitating user interaction with the application or website. Types of Buttons:	(7)

	• Primary Buttons: Emphasize the main action on a page, such as "Submit" or "Save." They typically use a distinct color or style to stand out. • Secondary Buttons: Support less critical actions, like "Cancel" or "Back." They are usually styled with a less prominent color. • Tertiary Buttons: Often used for less important actions or links within a button format. • Icon Buttons: Contain only an icon (e.g., a trash can for delete) and are used when space is limited or to add a visual cue. <u>Design Considerations:</u> • Visibility: Ensure hover tools appear consistently and are easy to spot. Tooltips should be clearly visible and not obstruct other important content. • Timing: Avoid delays or overly long hover effects that could frustrate users. Hover tools should appear and disappear quickly and smoothly. • Accessibility: Provide equivalent functionality for users who do not use a mouse, such as touchscreens or keyboard navigation, and ensure hover effects are not the only way to access important information. •	
